Q63: the deformation exponent j. Conjecture `conj:j` refuted entry by entry, and the closed form that replaces it

Clio

2 September 2026

Abstract

The 31 August paper classified every nonzero entry of $[e_i, R^{(\ell)}(t)]$ as either $\varepsilon q^k t^b(1+q^j t)$ (form (i)) or a t-monomial vanishing at $q=1$ (form (ii)), and conjectured (`conj:j`) that in form (i) $j = 1 + \big(\tau(\boldsymbol{\lambda},\gamma) - \tau(\nu,\gamma')\big)$, where τ is the cross-runner tie term of the level-ℓ telescope. Its whole support was that the two integers have the same range. We compare them entry by entry on all 1248 form-(i) entries of the original sweep.

`conj:j` **is false**: it holds on 678 of 1248 entries, against a floor of 598 scored by the constant $j \equiv 1$. It fails for a structural reason, visible in the smallest witness ($e=2$, $\ell=2$, $\boldsymbol{\lambda} = (\varnothing,(1))$): it subtracts two copies of the *same* statistic, and they cancel where they should not.

The refutation supplies the replacement. Every form-(i) entry has exactly two contributing paths; both lie in the *same* term of the commutator; their two nodes sit on one runner d at shifted contents c and $c+e$, where $c \to c+e-1$ is the net bead move. Writing τ for the tie term over runners below d and σ for the one over runners above,

$$\boxed{\; j \;=\; 1 + \varepsilon\big(\sigma(\boldsymbol{\lambda}; c, d) + \tau(\boldsymbol{\lambda}; c+e, d)\big), \qquad \varepsilon = +1 \iff c-1 \in M_d(\boldsymbol{\lambda}) \;}$$

with ε also the sign of the entry (Theorem 5.3). This is proved, not fitted: it follows from Theorem `thm:tele` of 31 August plus the fact that telescopes at contents c and $c+e$ differ by exactly one summand. `conj:j` is the special case in which one replaces σ by $-\tau$.

Two corollaries. The observed range $2-\ell \le j \le \ell$, previously an observation, follows: $|\sigma + \tau| \le \ell - 1$. And the same path analysis **proves the dichotomy** of Theorem `thm:rigid`, which stood as `computed`: an entry is of form (i) exactly when its two paths lie in the same term, and of form (ii) exactly when they do not — in which case their ribbon heights coincide and the two weights differ only in q, forcing the vanishing at $q=1$.

Verification: 1248/1248 in sample; 1189/1189 out of sample on six configurations absent from the 31 August sweep (including $\ell = 4$, $e = 5$, $|\boldsymbol{\lambda}| = 5$) and a depth-+6 control; and 1786/1786 for the intrinsic statement above, over thirteen configurations, which also exhibits $j = 4$ at $\ell = 4$ — a value outside every range seen before and inside the one now proved. Eight negative controls, each checked to be non-degenerate, all fail; the first negative control we ran was *not* non-degenerate and is reported as such in §7.

1 What was tested, and the two things that came out

The brief for this session was narrow and I want to record at the top that it was the right size. `conj:j` was the cheapest open node in the programme and carried the weakest evidence in it: two integers were known to have the same *range*, and had never been compared. Range agreement is a passing set, not a verification — the identical failure mode killed the $q = 1$ quantum-integer bridge on 31 August and hypothesis (H4) on 1 September. So: compute both integers on all 1248 entries, and look.

The comparison took under a minute of arithmetic and returned a refutation. The refutation then handed over the correct statement, because the quantity `conj:j` got wrong is a single, nameable term.

- §2: the independence check the brief asked for first.
- §3: the confusion matrix, the floor, and the first divergent entry in full, worked by hand.
- §4: the path census — why `conj:j` is not merely wrong but ill-posed as written.
- §5: Theorem 5.3, the closed form, with proof.
- §6: Theorem 6.1, the dichotomy, promoted from `computed` to `proved` by the same analysis.
- §7: verification and negative controls, including the one that was blind.

2 Conventions, and the independence check

Conventions are those of the 31 August paper (`2026-08-31-Q63-level-ell-telescope.tex`): rank $n = e \geq 2$, level $\ell \geq 1$, $N = e\ell$; $\boldsymbol{\lambda} = (\lambda^{(1)}, \ldots, \lambda^{(\ell)})$ with multicharge $\mathbf{s}$; runner-d Maya set M_d; shifted content x; the Chevalley action of Definition `def:chev` with tie-break $\prec$ on component indices; and the graded ribbon operator $R^{(\ell)}(t)|\boldsymbol{\lambda}\rangle = \sum t^h |\boldsymbol{\lambda}'\rangle$ of Definition `def:R`, the sum over single-runner bead moves $\kappa \mapsto \kappa + e$ with $h = \#\big(M_d \cap (\kappa, \kappa + e)\big)$.

In the implementation the Maya sets are indexed so that $M_d = \{\lambda_j^{(d)} - j + s_d\}$, one less than the paper's, and an addable node at (d, x) is $x - 1 \in M_d$, $x \notin M_d$. The two conventions agree on shifted contents; I use the implementation's throughout so that the formulas below can be read against the code.

The independence check. The 1 September session repaired a defect in the level-≥ 2 straightening of Uglov's Proposition 3.16, and its §Gaps asserted that `conj:j` is undamaged because $[e_i, R^{(\ell)}(t)]$ "uses neither the wedge nor the straightening". The brief asked that this be confirmed once rather than assumed, and it is confirmed: the module that computes $R^{(\ell)}(t)$, the Chevalley action and the commutator (`probes/2026-08-31-Q63-level-ell/fock_ell.py`) imports only `sympy` and `itertools`; R_e is built from the Maya sets directly; no symbol from the straightening route occurs anywhere in the dependency closure. The 1248 entries did not need regenerating, and were regenerated anyway — they came back identical (§7).

3 The comparison: `conj:j` is refuted

Conjecture 3.1 (`conj:j`, 31 August)**.** In form (i), $j = 1 + \big(\tau(\boldsymbol{\lambda}, \gamma) - \tau(\nu, \gamma')\big)$ for the two nodes involved.

Recall $\tau(\boldsymbol{\lambda},\gamma)=\sum_{d\prec d_0}\chi_d(x_0)$, the cross-runner tie term of Theorem `thm:tele`, where γ sits on runner d_0 at shifted content x_0, $\prec$ is "below" in the tie-break order, and $\chi_d(x)=[\text{addable at }(d,x)]-[\text{removable at }(d,x)]$.

Two remarks make the conjecture testable at all. First, τ reads only runners $d\neq d_0$; and for a form-(i) entry every multipartition occurring along either path ($\boldsymbol{\lambda}$, the intermediate, ν) agrees off runner d_0. *So the base is irrelevant* — $\tau(\boldsymbol{\lambda},\gamma)$, $\tau(\nu,\gamma)$ and $\tau(\text{mid},\gamma)$ are equal, verified 1248/1248. Second, the two nodes must be identified. The census of §4 shows the only available labelling is by t-degree: one node belongs to the path carrying t^b (call it γ_{lo}) and one to the path carrying t^{b+1} (γ_{hi}). Both orientations were tested.

prediction	agreement
$j=1+\tau(\gamma_{\text{lo}})-\tau(\gamma_{\text{hi}})$	678/1248
$j=1+\tau(\gamma_{\text{hi}})-\tau(\gamma_{\text{lo}})$	489/1248
$j\equiv 1$ (the level-1 answer; the floor)	598/1248
$\tau:=(\#\text{occupied runners}-1)$ (the rival)	598/1248

The better orientation of `conj:j` beats the constant $j\equiv 1$ by 80 entries out of 1248. Its confusion matrix is not concentrated on the diagonal:

obs j \ pred	-3	-2	-1	0	1	2	3	4	5
-1	1	0	7	2	7	0	0	0	0
0	0	2	23	59	100	4	0	0	0
1	0	0	13	84	**470**	27	4	0	0
2	0	0	0	15	205	**125**	23	0	0
3	0	0	0	0	39	17	**17**	3	1

`conj:j` also predicts values outside the observed range $(-3,-2,4,5)$, which $|\tau|\le\ell-1$ alone does not forbid once two τ's are subtracted.

3.1 The first divergent entry, in full

$e=2,\ \ell=2,\ \mathbf{s}=(0,0),\ i=1$	depth 14
$\boldsymbol{\lambda}=(\varnothing,(1))$	$\nu=((1),(1))$
$\Phi_{\nu\boldsymbol{\lambda}}(t)=q^{-3}\big(1+q^2t\big)$	so $\varepsilon=+1,\ k=-3,\ b=0,\ \ j=2$
`conj:j` predicts $1+\tau(\gamma_{\text{lo}})-\tau(\gamma_{\text{hi}})=1+1-1=1$	Theorem 5.3 predicts 2

This is small enough to check by hand, and the hand check is where the mechanism becomes visible. In the implementation's indexing, $M_0=\{-1,-2,-3,\dots\}$ and $M_1=\{0,-2,-3,\dots\}$. The odd-content ($i=1$) nodes of $\nu=((1),(1))$, whose Maya sets are both $\{0,-2,-3,\dots\}$, are addable at $x=1$ and $x=-1$ on each of the two runners. With the tie-break $\prec$ of Definition `def:chev` (smaller runner index ranks higher), counting addable minus removable strictly below:

$$N^{<}(\nu,(1,d{=}0))=\underbrace{1}_{(-1,0)}+\underbrace{1}_{(-1,1)}+\underbrace{1}_{(1,1)}=3,\qquad N^{<}(\nu,(-1,d{=}0))=\underbrace{1}_{(-1,1)}=1.$$

The two paths are: ribbon $-1\mapsto 1$ on runner 0 then removal at $x=1$ (height 0, weight q^{-3}), and ribbon $-2\mapsto 0$ then removal at $x=-1$ (height 1, weight q^{-1}). Hence $\Phi=q^{-3}+q^{-1}t=q^{-3}(1+q^2t)$ and $j=3-1=2$.

Now the two τ's. Runner 0 is above runner 1, so τ at a node of runner 0 is χ_1 at that content. Both $x = 1$ and $x = -1$ are addable on runner 1, so

$$\tau(\gamma_{\mathrm{lo}}) = \chi_1(1) = +1, \qquad \tau(\gamma_{\mathrm{hi}}) = \chi_1(-1) = +1,$$

and `conj:j` returns $1 + 1 - 1 = 1$. *The two tie terms cancelled.* They cancel because `conj:j` subtracts one copy of the same statistic from another. What the truth requires here is τ at the low node and σ — the tie term over runners *above* — at the high node; runner 0 is the top runner, so $\sigma = 0$, and $j = 1 + 0 + 1 = 2$.

Remark 3.2. This is a cleaner death than a numerical mismatch. `conj:j` is not off by a correction term; it pairs the wrong two statistics, and the pairing it chooses is exactly the one that cancels. That it still scores 678 is not partial credit: it scores 678 because $\sigma = \tau = 0$ on the 598 entries where nothing is happening, plus 80 coincidences.

4 Why it is ill-posed: the path census

`conj:j` writes $\tau(\boldsymbol{\lambda}, \gamma) - \tau(\nu, \gamma')$, which presumes one node lives in $\boldsymbol{\lambda}$ and one in ν — that is, that the two paths come from the two different terms e_iR and Re_i of the commutator. They never do.

Proposition 4.1 (Census)**.** *Over all* 1248 *form-(i) entries of the 31 August sweep: every entry has exactly two contributing paths; in every entry both paths lie in the same term (*1204 *in e_iR,* 44 *in Re_i); in every path the ribbon and the node sit on the same runner; the two nodes sit on the same runner as each other; and their shifted contents differ by exactly e.*

The same-term fact is forced, and worth stating as such: a form-(i) entry $\varepsilon q^k t^b(1 + q^j t)$ has *both* t-coefficients of sign ε, whereas the two terms of a commutator enter with opposite signs. So a two-path entry drawn from opposite terms cannot be of form (i). What the census adds is that form-(i) entries never have more than two paths, so no cancellation restores the possibility.

Lemma 4.2 (Same runner)**.** *Every path contributing to a form-(i) entry has its ribbon and its node on the same runner.*

Proof. A path with ribbon on runner d_1 and node on $d_2 \neq d_1$ changes component d_1 (by $+e$ boxes) and component d_2 (by -1 box), so its target differs from $\boldsymbol{\lambda}$ in two components. Form-(i) entries are by definition those where ν and $\boldsymbol{\lambda}$ differ in a single component. If instead $d_1 = d_2 = d$ then component d changes by $e - 1 \neq 0$ boxes ($e \geq 2$), so it is exactly component d that differs. □

5 The closed form

Fix a form-(i) entry. By Lemma 4.2 all activity is on one runner d; write $M = M_d(\boldsymbol{\lambda})$ for its Maya set. The net change of M is one bead moving from c to $c + e - 1$ (Lemma 5.2 below), and $c \equiv i \pmod{e}$.

Definition 5.1. For $x \in \mathbb{Z}$ put, all χ's read in $\boldsymbol{\lambda}$,

$$\tau(\boldsymbol{\lambda}; x, d) = \sum_{d' \prec d} \chi_{d'}(x), \qquad \sigma(\boldsymbol{\lambda}; x, d) = \sum_{d' \succ d} \chi_{d'}(x),$$

the tie terms over runners below and above d. Thus $\tau+\sigma+\chi_d(x)=\sum_{d'}\chi_{d'}(x)$, and τ is the τ of Theorem `thm:tele`. Both are unchanged if $\boldsymbol{\lambda}$ is replaced by any multipartition agreeing with it off runner d.

Lemma 5.2 (Exactly two paths)**.** *Let ν be obtained from $\boldsymbol{\lambda}$ by the single bead move $c\mapsto c+e-1$ on runner d. Then $[e_i,R^{(\ell)}(t)]$ has exactly two paths $\boldsymbol{\lambda}\to\nu$, namely*

$$\begin{aligned}\textit{the } c\textit{-path:}\quad & \textit{ribbon } c-1\mapsto c+e-1,\ \textit{node at content } c;\\ \textit{the } (c+e)\textit{-path:}\quad & \textit{ribbon } c\mapsto c+e,\ \textit{node at content } c+e;\end{aligned}$$

each occurring in exactly one of the two terms, according to

$$c\textit{-path lies in } e_iR \iff c-1\in M,\qquad (c+e)\textit{-path lies in } e_iR \iff c+e\notin M.$$

Their ribbon heights satisfy

$$h_c-h_{c+e}=\begin{cases}+1 & \textit{both paths in } e_iR,\\ -1 & \textit{both paths in } Re_i,\\ 0 & \textit{otherwise.}\end{cases}$$

Proof. A path is a ribbon $b\mapsto b+e$ and a node move $p\mapsto p-1$, in one of the two orders. Composing them, the net change of M is a single bead move only if the two moves share an endpoint: either $p=b+e$, giving $b\mapsto b+e-1$, or $p-1=b$, giving $b+1\mapsto b+e$. Setting $c=b$ in the first and $c=b+1$ in the second gives exactly the two displayed paths, both of displacement $e-1$; every other pair (b,p) leaves two beads moved. This is the whole list.

Legality. Note $c\in M$ and $c+e-1\notin M$, since the net move takes the bead at c to the free site $c+e-1$. For the $(c+e)$-path in the order e_iR: the ribbon $c\mapsto c+e$ needs $c+e\notin M$; the subsequent removal at $c+e$ then needs $c+e\in M'$ (true, just placed) and $c+e-1\notin M'$ (true). In the order Re_i: the removal at $c+e$ needs $c+e\in M$ and $c+e-1\notin M$; the ribbon $c\mapsto c+e$ then needs $c+e\notin M''$ (true, just vacated). So exactly one of the two orders is legal, selected by whether $c+e\in M$. Identically, the c-path in the order e_iR needs $c-1\in M$ for the ribbon $c-1\mapsto c+e-1$, and in the order Re_i needs $c-1\notin M$ for the removal at c; exactly one is legal.

Heights. Put $H=\#\big(M\cap\{c+1,\dots,c+e-2\}\big)$ and recall $c\in M$, $c+e-1\notin M$. For the $(c+e)$-path in e_iR the ribbon acts on M: $h_{c+e}=\#\big(M\cap(c,c+e)\big)=H$. In Re_i it acts on $M''=M\setminus\{c+e\}\cup\{c+e-1\}$, so $h_{c+e}=H+1$. For the c-path in e_iR the ribbon acts on M over the window $(c-1,c+e-1)$: $h_c=\#\big(M\cap\{c,\dots,c+e-2\}\big)=H+1$. In Re_i it acts on $M'=M\setminus\{c\}\cup\{c-1\}$, so $h_c=H$. The four combinations give the three stated values. □

Theorem 5.3 (Closed form for j)**.** *Let $\Phi_{\nu\boldsymbol{\lambda}}(t)$ be a form-(i) entry, with net runner-d bead move $c\mapsto c+e-1$ and $M=M_d(\boldsymbol{\lambda})$. Then*

$$\Phi_{\nu\boldsymbol{\lambda}}(t)=\varepsilon\, q^kt^b\big(1+q^jt\big),\qquad \varepsilon=\begin{cases}+1, & c-1\in M,\\ -1, & c-1\notin M,\end{cases}$$

and

$$\boxed{j\ =\ 1+\varepsilon\Big(\sigma(\boldsymbol{\lambda};c,d)+\tau(\boldsymbol{\lambda};c+e,d)\Big).}$$

Consequently $|j-1|\le\ell-1$, i.e. $2-\ell\le j\le\ell$.

Proof. By Proposition 4.1 and Lemma 5.2 the entry is form (i) exactly when the two paths lie in the same term, i.e. when $c-1 \in M$ and $c+e \notin M$ (both in e_iR, sign $+1$), or $c-1 \notin M$ and $c+e \in M$ (both in Re_i, sign -1); so ε is as stated. Write N_c, N_{c+e} for the Chevalley exponents of the two paths, each weight being q^{-N} read on that path's own base.

Case $\varepsilon = +1$. Here $h_c = h_{c+e}+1$, so the $(c+e)$-path carries t^b and $\Phi = q^{-N_{c+e}}t^b\big(1+q^{N_{c+e}-N_c}t\big)$, i.e. $j = N_{c+e} - N_c$. Both weights are read on the same base ν (in the order e_iR the node is removed last, and its target is ν). Put $A_{d'}(x) = \sum_{k\geq 1}\chi^{\nu}_{d'}(x-ke)$ and $T(x) = \sum_{d'} A_{d'}(x)$. Theorem `thm:tele` says $N^{<}(\nu,\gamma) = T(x)+\tau(x,d)$ for a node γ at (d,x). Since $A_{d'}(x+e) = \chi^{\nu}_{d'}(x) + A_{d'}(x)$, the two telescopes differ by exactly one summand: $T(c+e)-T(c) = \sum_{d'}\chi^{\nu}_{d'}(c)$. Hence

$$j = \Big(\textstyle\sum_{d'}\chi^{\nu}_{d'}(c)\Big) + \tau(c+e,d) - \tau(c,d) = \chi^{\nu}_{d}(c) + \sigma(c,d) + \tau(c+e,d),$$

using $\sum_{d'} = \chi_d + \tau + \sigma$ to cancel $\tau(c,d)$. Finally $\chi^{\nu}_d(c) = +1$: the c-path's node is addable at (d,c) in its target ν.

Case $\varepsilon = -1$. Here $h_c = h_{c+e} - 1$, so the c-path carries t^b and $j = N_c - N_{c+e}$. The two weights are read on different bases, $M' = M\setminus\{c\}\cup\{c-1\}$ and $M'' = M\setminus\{c+e\}\cup\{c+e-1\}$, which agree with M — and with each other — off runner d; so all τ and σ terms may be read in $\boldsymbol{\lambda}$. Off runner d, using $A_{d'}(c) = A_{d'}(c+e) - \chi_{d'}(c)$,

$$\sum_{d'\neq d}\big(A_{d'}(c) - A_{d'}(c+e)\big) = -\sum_{d'\neq d}\chi_{d'}(c) = -\tau(c,d) - \sigma(c,d).$$

On runner d: the arguments of $A^{M'}_d(c)$ are all $\leq c-e \leq c-2$, and M' differs from M only at $c-1$ and c, so $A^{M'}_d(c) = A^{M}_d(c)$; the arguments of $A^{M''}_d(c+e)$ are all $\leq c$, and M'' differs from M only at $c+e-1$ and $c+e$, so $A^{M''}_d(c+e) = A^{M}_d(c+e)$. Hence the runner-$d$ contribution is $A^M_d(c) - A^M_d(c+e) = -\chi^{\boldsymbol{\lambda}}_d(c) = +1$, because in the case $\varepsilon=-1$ the c-path's node is *removable* at (d,c) in $\boldsymbol{\lambda}$. Adding the tie terms $\tau(c,d)-\tau(c+e,d)$ gives

$$j = 1-\tau(c,d)-\sigma(c,d)+\tau(c,d)-\tau(c+e,d) = 1-\sigma(c,d)-\tau(c+e,d),$$

which is the boxed formula with $\varepsilon = -1$.

The range: $\sigma(c,d)$ is a sum of $\chi_{d'}(c)\in\{-1,0,1\}$ over the runners above d and $\tau(c+e,d)$ a sum over those below, so $|\sigma+\tau|$ is at most the number of runners other than d, namely $\ell-1$. ☐

Remark 5.4 (What `conj:j` got right and wrong). `conj:j` is the assertion that the telescope difference $T(c+e)-T(c)$ equals 1. It is $1+\sum_{d'\neq d}\chi_{d'}(c)$, and the extra sum is exactly what `conj:j` misses. This is visible in the data: with $\Delta T := T(c+e)-T(c)$ one has $j = \Delta T + \Delta\tau$ identically on all 1248 entries, $\Delta T = 1$ on exactly 678 of them, and those 678 are precisely the entries on which `conj:j` holds — the same set, not merely the same count. So the diagnosis is exact: `conj:j` is true iff the level-1 telescope difference survives, and it survives just over half the time.

Remark 5.5 (The shape of the answer). $j-1$ is a sum of two tie terms of *opposite orientation* evaluated at *different contents*: the above-term at c and the below-term at $c+e$. That asymmetry is not decoration. The two nodes of a form-(i) entry are the two ends of one ribbon window; the merge-by-content order sees the lower end from above and the upper end from below, and j is the total cross-runner traffic across the window. At $\ell = 1$ there is no traffic and $j=1$, which is the Q59 theorem.

6 The dichotomy, promoted to a theorem

Theorem `thm:rigid` of 31 August — that every nonzero entry is of form (i) or form (ii) — was recorded as `computed`, with no proof. The path analysis proves it.

Theorem 6.1 (Dichotomy)**.** *Every nonzero entry $\Phi_{\nu\boldsymbol{\lambda}}(t)$ of $[e_i, R^{(\ell)}(t)]$ has exactly two contributing paths, and*

(i) if both lie in the same term of the commutator, then ν and $\boldsymbol{\lambda}$ differ in one component, the two ribbon heights differ by 1, and $\Phi = \varepsilon q^k t^b(1 + q^j t)$ with j as in Theorem 5.3;

(ii) otherwise the two ribbon heights coincide and $\Phi = t^h\big(q^{-N_1} - q^{-N_2}\big)$, a t-monomial vanishing at $q = 1$.

Proof. Fix ν with $\Phi_{\nu\boldsymbol{\lambda}} \neq 0$ and let Δ be the net change of Maya sets.

If Δ moves beads on two runners $d_1 \neq d_2$, every path consists of a ribbon on d_1 and a node move on d_2 (Lemma 4.2's computation run backwards), and the pair (b, p) is determined by Δ; the two orders e_iR and Re_i are both legal, since moves on different runners are independent. So there are exactly two paths, in opposite terms, and the ribbon height is the same in both — the node move is on another runner and does not meet the ribbon window. Case (ii).

If Δ moves two beads on one runner d, say $b \mapsto b+e$ and $p \mapsto p-1$ with $p \neq b+e$ and $p-1 \neq b$, then again (b, p) is determined and both orders are legal, so there are exactly two paths in opposite terms. The node move changes $\#\big(M \cap (b, b+e)\big)$ only if exactly one of $p-1, p$ lies in $(b, b+e)$, i.e. only if $p = b+e$ or $p-1 = b$, both excluded; so the heights agree. Case (ii).

If Δ moves one bead on one runner, Lemma 5.2 applies: exactly two paths, the c-path and the $(c+e)$-path, each in a term determined by a local bead condition, with $h_c - h_{c+e} \in \{+1, -1, 0\}$ according to whether the terms agree (± 1) or not (0). If they agree, the two monomials have equal signs and adjacent t-degrees: case (i), and Theorem 5.3 computes j. If they disagree, the signs are opposite and the degrees equal, giving $t^h(q^{-N_1} - q^{-N_2})$: case (ii). (If $N_1 = N_2$ the entry is zero and does not arise.) ☐

Corollary 6.2. *Form (ii) is empty at $\ell = 1$ and $j \equiv 1$ there.*

Proof. At $\ell = 1$ there are no two-runner entries; the one-runner two-bead case and the opposite-term case do occur in principle, but $\sigma = \tau = 0$, so where form (i) holds $j = 1$. The claim that form (ii) is empty at $\ell = 1$ is the observation of Remark `rem:l1` (350 entries, all of form (i)) and is *not* proved here: Theorem 6.1 permits form-(ii) entries at $\ell = 1$ and the sweep found none. This is recorded as a gap in §8. ☐

7 Verification and negative controls

In sample. All 1248 form-(i) entries over the nine triples $(e, \ell, \mathbf{s})$ of Remark `rem:verif-rigid`, all multipartitions of size ≤ 4, all i. The count, the j-distribution $\{-1{:}17,\ 0{:}188,\ 1{:}598,\ 2{:}368,\ 3{:}77\}$ and the range $[2-\ell, \ell]$ all reproduce the 31 August sweep, which is the check that this session's pipeline computes the same object. Theorem 5.3: 1248/1248.

Out of sample. Six configurations absent from the 31 August sweep — $(e, \ell, \mathbf{s}) = (2, 2, (1, 0))$ with $|\boldsymbol{\lambda}| \le 4$; $(3, 2, (0, 2))$; $(2, 4, (0, 0, 0, 0))$; $(5, 2, (0, 0))$; $(3, 3, (0, 1, 2))$; and $(2, 2, (0, 0))$ pushed to $|\boldsymbol{\lambda}| \le 5$ — giving 1189 further form-(i) entries. Theorem 5.3: 1189/1189. These reach $\ell = 4$ and $e = 5$, both new.

The intrinsic statement. The two runs above test the formula in its path-level form, reading σ and τ on each path's own weight base. Theorem 5.3 is stated intrinsically instead: c is read off Δ, ε off the single bead condition $c - 1 \in M$, and all χ's in $\boldsymbol{\lambda}$ itself. That version was tested separately over thirteen configurations (the nine in-sample triples plus $(2, 2, (1, 0))$ at $|\boldsymbol{\lambda}| \le 5$, $(5, 2, (0, 0))$, $(2, 4, (0, 0, 0, 0))$ and $(3, 3, (0, 1, 2))$), giving 1786 form-(i) entries. All four of its component claims hold 1786/1786: $\varepsilon = +1 \iff c - 1 \in M$; form (i) $\iff \big(c + e \in M \iff c - 1 \notin M\big)$; the boxed formula; and $|\sigma + \tau| \le \ell - 1$. The $\ell = 4$ configuration produced 8 entries with $j = 4$, a value no previous sweep had seen, predicted correctly.

Depth. The brief's standing instruction is to check the depth first on any failure. There were no failures, so the control was run positively instead: the triple $(2, 3, (0, 0, 0))$ was recomputed with the abacus depth raised by 6. It returned the same 201 form-(i) entries with the same j's and 0 form-(ii)/unclassified drift, so the 1248 are not truncation artefacts.

Negative controls. *The first one we ran was blind, and that is the most useful thing in this section.* The brief specified: "perturb τ by $+1$ on one runner and confirm the comparison goes red". It does not go red. It scores 678/1248 with a bit-identical confusion matrix, because the formula uses τ only through a difference of two nodes on the *same* runner d, so any perturbation depending on (d', d) alone cancels identically. A control that cannot move the score verifies nothing. Every control below was therefore first checked for non-degeneracy — it must change at least one prediction — and the count of moved predictions is reported.

variant	score	predictions moved
Theorem 5.3	**1248/1248**	—
`conj:j` as written	678/1248	570
tie-break order reversed ($\sigma \leftrightarrow \tau$)	500/1248	748
σ, τ evaluated at the other node's content	500/1248	748
the σ term dropped	892/1248	356
the τ term dropped	892/1248	356
the ε sign branch ignored	1225/1248	23
rival: $\tau :=$ (#occupied runners − 1)	598/1248	650
constant $j \equiv 1$ (the floor)	598/1248	650

Planted errors. The comparison was also tested for the ability to see a failure at all: perturbing 1, 5 and 20 of the *observed* j's by $+1$ drops the score to 1247, 1243 and 1228 respectively — caught one for one.

What else would pass. Stating this before concluding, as the brief requires. The formula has no fitted parameter: σ and τ are the two halves of the tie term of Theorem `thm:tele`, the contents c and $c + e$ are forced by Lemma 5.2, and the leading 1 is the runner-d term $\chi_d = \pm 1$, proved rather than chosen. The one place a coincidence could hide is the sign branch ε: ignoring it still scores 1225/1248, because only 44 of the 1248 entries lie in Re_i and 21 of those have $\sigma + \tau = 0$. The branch is therefore supported by 23 entries in sample — thin — which is why the out-of-sample sweep was run and why the $\varepsilon = -1$ case is given a separate proof in Theorem 5.3 rather than inferred.

8 Gaps

1. **Form (ii) is not computed.** Theorem 6.1 shows every form-(ii) entry is $t^h(q^{-N_1} - q^{-N_2})$ and hence vanishes at $q = 1$, but gives no closed form for $N_1 - N_2$. The same σ/τ analysis should apply and was not attempted; this is the obvious next half-session.

2. **Form (ii) is not proved empty at $\ell = 1$.** Theorem 6.1 permits one-runner two-bead entries and opposite-term entries at $\ell = 1$; the $\ell = 1$ sweep (350 entries) found none. Either they are excluded by a positivity or an interval argument I have not found, or they exist beyond $|\lambda| \leq 7$.

3. **Sizes.** $|\boldsymbol{\lambda}| \leq 5$, bead depths 12–20. Larger than every previous sweep, still small. In particular $\ell \leq 4$: the tie terms σ and τ are sums over $\ell - 1$ runners and have been exercised with at most three of them.

4. **One convention.** Everything is computed at tie-break $+1$ (smaller runner index ranks higher), the convention pinned in Remark `rem:l1`. Under the opposite tie-break σ and τ exchange roles, which is the "tie-break order reversed" control above; the formula is not tie-break-invariant, and that is correct rather than alarming, but the mirror statement was not written out.

5. **Nothing here touches Proposition `prop:heis`.** The obstruction that $R^{(\ell)}(-q^{-1}) \neq B^{[e]}_{-1}$ for $\ell \geq 2$ stands untouched. Theorem 5.3 describes the commutator of the graded ribbon operator, which after `prop:heis` is no longer known to be the representation-theoretically meaningful one. The motivation gap named in the 1 September brief is not closed by this session.

9 Corrections to the 31 August paper

1. Conjecture `conj:j` is **refuted** (678/1248) and replaced by Theorem 5.3. Its supporting sentence — "the evidence is the range" — was an accurate description of the evidence and the evidence was insufficient.

2. Theorem `thm:rigid`, recorded as `computed`, is **proved** (Theorem 6.1), except for the claim that form (ii) is empty at $\ell = 1$, which remains computational (§8).

3. The observed range $2 - \ell \leq j \leq \ell$ of `thm:rigid` is **proved**, as a corollary of Theorem 5.3.

4. The closing sentence of §`sec:tele` — "if `conj:j` holds then §`sec:tele` and §`sec:rigid` are one statement" — **stands, with the conclusion reached by another route**: they are one statement, because $j - 1$ is built from the two halves of the same tie term. The bridge is real; the formula proposed for it was wrong.